

# KUO-HAN HUNG

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## Education

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### Stanford University

June 2027 (Expected)

*M.S. in Computer Science*

*Stanford, CA*

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

### National Taiwan University

Dec. 2024

*B.S. in Computer Science and Information Engineering*

*Taipei, Taiwan*

- GPA: 4.24/4.3.

## Technical Skills

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- **Robotics:** Real-Robot Deployment and Training, Real-to-Sim, Real-Time Control, Manipulation, ROS.
- **Parallel Computing:** Distributed Training, Ray, Multiprocessing/Asyncio, Vectorized Environments.
- **Infrastructure:** Linux, Docker, SLURM, Git, Azure.
- **Languages & Frameworks:** Python, C/C++, Golang, JavaScript, SQL, Bash, PyTorch, JAX, NumPy.

## Research Experience

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### Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

*Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn*

*Stanford, CA*

- **Visual representation pretraining** for reinforcement learning (ongoing).
- **RL finetuning of real-time VLA policies** under inference latency. [1]
  - Decoupled slow VLA action proposals from a fast edit policy that reacts to the latest observation.
  - Raised success rate from 42% to 97% on four dynamic real-world tasks with 10 min of robot data.
- **Sample-efficient RL finetuning of pretrained VLAs** in the real world. [2]
  - Built a stable RL finetuning system with residual action edits and Q-guided sampling.
  - Reached 30/30 success on all tasks with 19 min of online data on average; code open-sourced.
- **Scaling Q-functions** with Transformers in offline RL. [3]
  - Introduced attention entropy regularization to stabilize and scale Transformer value functions.
  - Showed that prior methods fail to scale, while TQL gains 43% as model size grows.

### Embodied AI Lab

June 2025 – Sept. 2025

*Undergraduate Researcher, Advisor: Prof. Tsung-Wei Ke*

*Taipei, Taiwan*

- **Generalized dexterous manipulation policies** via digital twins.
  - Built a pipeline to convert videos into digital twins using mesh generation and object/human pose estimation.
  - Trained residual RL policies on digital twins from in-the-wild videos, yielding diverse dexterous behaviors.

### Robot Learning Lab

Oct. 2024 – May 2025

*Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun*

*Taipei, Taiwan*

- **Manipulation policies learned from action-free videos.** [5]
  - Designed a Transformer to learn skills from action-free videos, using optical flow as an action surrogate.
  - Enabled multi-task and long-horizon planning by translating flow-based skill plans into executable actions.

### Communication and Multimedia Lab

May 2022 – May 2024

*Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen*

*Taipei, Taiwan*

- **Vision-instruction correlation rewards** for long-horizon manipulation. [4]
  - Finetuned LLMs and VLMs into a hierarchical video reward model for generating stage-aware signals.
  - Achieved a 43% improvement in success rate over state-of-the-art baselines on long-horizon tasks.
- **Behavior error detection** in few-shot imitation policies via representation patterns. [7]

## Work Experience

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### IBM Research

June 2024 – Sept. 2024

Research Intern, Host: Dr. Pin-Yu Chen

Yorktown Heights, NY

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents**. [6]
- Traced the effect to specific attention heads using mechanistic interpretability.
- Designed an attention-based detector, a 31% AUROC improvement over the prior state of the art.

### Microsoft

July 2023 – June 2024

Applied Scientist Intern, Host: Dr. Chiquan Zhang

Mountain View, CA (Remote)

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra**.
- Developed an LLM-powered search system that adapts to user intent and geographic context.

## Selected Publications

(\* indicates equal contribution)

- [1] **Reinforcement Learning for Real-Time Vision-Language-Action Policies**  
Perry Dong\*, **Kuo-Han Hung\***, Dorsa Sadigh, and Chelsea Finn  
Under Review
- [2] **EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models**  
Perry Dong\*, **Kuo-Han Hung\***, Tian Gao, Dorsa Sadigh, and Chelsea Finn  
Conference on Robot Learning (CoRL), 2026
- [3] **TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse**  
Perry Dong\*, **Kuo-Han Hung\***, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn  
International Conference on Machine Learning (ICML), 2026
- [4] **VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation**  
**Kuo-Han Hung\***, Pang-Chi Lo\*, Jia-Fong Yeh\*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu  
International Conference on Learning Representations (ICLR), 2025
- [5] **Learning Skill Abstraction from Action-Free Videos via Optical Flow**  
Hung-Chieh Fang\*, **Kuo-Han Hung\***, Chu-Rong Chen, ..., Min-Hung Chen, and Shao-Hua Sun  
ICML Workshop on Building Physically Plausible World Models, 2025
- [6] **Attention Tracker: Detecting Prompt Injection Attacks in LLMs**  
**Kuo-Han Hung**, Ching-Yun Ko, Ambrish Rawat, I-Hsin Chung, Winston Hsu, and Pin-Yu Chen  
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), 2025
- [7] **AED: Adaptable Error Detection for Few-shot Imitation Policy**  
Jia-Fong Yeh, **Kuo-Han Hung**, Pang-Chi Lo, Chi-Ming Chung, ..., Yi-Ting Chen, and Winston Hsu  
Conference on Neural Information Processing Systems (NeurIPS), 2024

## Awards & Honors

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- **Bachelor Thesis Dean's Award**, National Taiwan University — top 2 theses in the department. May 2024
- **Presidential Award**, National Taiwan University — top 2% of the academic year. Sept. 2022, Sept. 2023
- **Dean's List**, National Taiwan University — top 5% of the academic year. June 2021, 2022, 2023